

Supplementary methods and complete results

Reassessing Pruning and Feature Distillation for Lightweight YOLOv8

Hyerim Choi and Tae-Kook Kim

This supplement reports every prespecified seed and frozen external model. It complements the main article with full numerical outcomes, protocol boundaries and artifact identifiers. AP values are fractions; paired absolute differences are percentage points (pp). Earlier and fresh phases are never pooled as independently trained starting-model repetitions.

Table S1 All frozen external outcomes

All 25 checkpoints were fixed before inference on the 7,481 publicly labeled KITTI training-partition images. No training, adaptation or selection used these scores. Car and Pedestrian use native ignore and difficulty rules, IoU thresholds 0.7 and 0.5, and AP_R40. E, M and H mean easy, moderate and hard. The prespecified external summary is the mean of the two moderate values; this is not BDD ten-class AP or an official hidden-test result.

Model	Car E	Car M	Car H	Ped. E	Ped. M	Ped. H
Archived nano	0.890393	0.782833	0.642672	0.733101	0.648050	0.574283
Archived r0.15	0.902498	0.773002	0.635660	0.704901	0.618611	0.550164
Archived r0.30	0.884439	0.764299	0.626181	0.627596	0.541783	0.481454
Archived r0.45	0.893042	0.754823	0.617756	0.609457	0.526388	0.467659
Archived recovery	0.892955	0.755653	0.618667	0.601499	0.512478	0.453077
Earlier control s0	0.885975	0.776713	0.639185	0.731045	0.645224	0.572086
Earlier control s1	0.886381	0.777152	0.639476	0.731898	0.644400	0.571139
Earlier control s2	0.886831	0.777143	0.639785	0.732008	0.643331	0.569996
Earlier MSE s0	0.888801	0.778016	0.640202	0.730243	0.644205	0.570786
Earlier MSE s1	0.887134	0.777821	0.639855	0.728831	0.641314	0.567605
Earlier MSE s2	0.887327	0.777175	0.639789	0.731753	0.643390	0.570147
Earlier CWD s0	0.889021	0.778460	0.641197	0.730376	0.643974	0.569774
Earlier CWD s1	0.889291	0.778640	0.640286	0.729265	0.642356	0.567733
Earlier CWD s2	0.887490	0.776800	0.640013	0.730685	0.642589	0.568966
Fresh nano	0.886840	0.779739	0.641362	0.724036	0.636483	0.562000
Fresh large	0.883933	0.796329	0.656186	0.820446	0.738980	0.654862
Fresh control s0	0.884520	0.775850	0.637928	0.715649	0.628682	0.554569
Fresh control s1	0.884141	0.775497	0.637153	0.715069	0.626563	0.553678
Fresh control s2	0.885164	0.776161	0.637910	0.714121	0.627200	0.553737
Fresh MSE s0	0.883689	0.775760	0.637208	0.716664	0.628929	0.554917
Fresh MSE s1	0.884409	0.775788	0.637240	0.715325	0.627562	0.555164

Model	Car E	Car M	Car H	Ped. E	Ped. M	Ped. H
Fresh MSE s2	0.882628	0.776020	0.637311	0.715111	0.628117	0.554680
Fresh CWD s0	0.886319	0.775565	0.637090	0.717314	0.629118	0.555334
Fresh CWD s1	0.885176	0.775961	0.637386	0.715471	0.628026	0.554180
Fresh CWD s2	0.883530	0.777752	0.638380	0.720980	0.632023	0.557072

All 150 class/difficulty AP values independently match the means of native precision positions 1–40 (position 0 excluded). An additional file audit rehashed all 25 source checkpoints, 187,025 serialized predictions and 23,400 batch markers. The archive contents were independently rehashed after transfer. The original native evaluator and local patch, eight synthetic test cases and frozen roster are in the repository.

S2 Every matched continuation endpoint

Each phase contains control, MSE and CWD with data/augmentation seeds 0/1/2, using one shared trained student and teacher per phase. The earlier phase uses the original 70,000-image conversion; the fresh phase uses 69,863 reconstructed-source images with fresh COCO initialization. Primary validation is FP32, batch 32, all ten classes. Every endpoint is final epoch 20 EMA, not the best epoch. $\Delta 95$ is AP50:95 minus the same-seed control. No objective-specific coefficient search or convergence claim is made.

Phase	Arm	Seed	AP50	AP50:95	$\Delta 95$ pp	AMP skips
Earlier	CONTROL	0	0.463525	0.258903	+0.00000	5
Earlier	CONTROL	1	0.463442	0.259059	+0.00000	5
Earlier	CONTROL	2	0.464460	0.259402	+0.00000	4
Earlier	MSE	0	0.463549	0.258554	-0.03485	4
Earlier	MSE	1	0.463120	0.259081	+0.00226	6
Earlier	MSE	2	0.464154	0.258976	-0.04266	5
Earlier	CWD	0	0.463270	0.258839	-0.00642	4
Earlier	CWD	1	0.462477	0.258787	-0.02721	5
Earlier	CWD	2	0.464348	0.259053	-0.03496	4
Fresh	CONTROL	0	0.464381	0.258917	+0.00000	5
Fresh	CONTROL	1	0.464164	0.259107	+0.00000	5
Fresh	CONTROL	2	0.464488	0.259347	+0.00000	5
Fresh	MSE	0	0.464083	0.258581	-0.03354	5
Fresh	MSE	1	0.463951	0.258756	-0.03517	5
Fresh	MSE	2	0.464486	0.259076	-0.02708	4
Fresh	CWD	0	0.463050	0.258131	-0.07854	5
Fresh	CWD	1	0.463752	0.258700	-0.04075	4

Phase	Arm	Seed	AP50	AP50:95	$\Delta 95$ pp	AMP skips
Fresh	CWD	2	0.462992	0.258187	-0.11594	5

Earlier runs attempt 21,880 minibatches; fresh runs attempt 21,840. Applied updates equal attempted minibatches minus the listed skips, and each EMA counter agrees. Every completed run changes 183 trainable student tensors while preserving the DFL projection. Final student, adapter, EMA and optimizer tensors and objective means are finite. Raw nonfinite epoch gradient-norm diagnostics are retained; they occur in epochs with AMP-skipped updates. The teacher is frozen with detached targets by executed code; no independent per-run teacher-gradient field is fabricated.

S3 Complete paired external differences

Phase	Arm	Seed 0 pp	Seed 1 pp	Seed 2 pp	Mean pp	SD pp
Earlier	MSE	+0.01415	-0.12081	+0.00452	-0.03405	0.07529
Earlier	CWD	+0.02482	-0.02777	-0.05420	-0.01905	0.04023
Fresh	MSE	+0.00788	+0.06447	+0.03880	+0.03705	0.02833
Fresh	CWD	+0.00755	+0.09634	+0.32072	+0.14154	0.16140

Fresh external paired differences are positive for every seed, but all nine continuations score below fresh nano 0.708111 on the external summary. Dataset distribution, category set and AP definition change jointly between BDD and KITTI. The sign contrast cannot isolate a domain-shift mechanism or establish a generally superior KD method.

S4 Reconstructed-label evaluation of all 27 models

These secondary evaluations all use reconstructed legacy validation labels, FP32 and batch 8. Global values average ten classes over 10,000 images. Day/night/dawn values average the fixed nine road classes, excluding train: 5,258 daytime, 3,929 night and 778 dawn/dusk images. The remaining 35 undefined images have incomplete class support and remain in global evaluation. Day, night and dawn columns below are AP50:95, not AP50.

Model	All AP50	All AP95	Day AP95	Night AP95	Dawn AP95
Earlier control s0	0.463689	0.258936	0.293582	0.275155	0.291148
Earlier control s1	0.463595	0.259110	0.293761	0.276094	0.287872
Earlier control s2	0.464487	0.259430	0.293903	0.276998	0.289226
Earlier CWD s0	0.463290	0.258887	0.293596	0.275214	0.290380
Earlier CWD s1	0.462459	0.258880	0.293518	0.275652	0.289250
Earlier CWD s2	0.464457	0.259140	0.294122	0.275587	0.289317
Earlier MSE s0	0.463522	0.258552	0.293208	0.275210	0.289144
Earlier MSE s1	0.463069	0.259041	0.294007	0.275627	0.287673
Earlier MSE s2	0.464153	0.258969	0.293798	0.276174	0.287772
Archived large	0.583273	0.342325	0.391966	0.353435	0.381286
Archived nano	0.463537	0.259514	0.294559	0.275343	0.289410

Model	All AP50	All AP95	Day AP95	Night AP95	Dawn AP95
Archived r0.30	0.380617	0.208752	0.238073	0.218778	0.227608
Archived r0.15	0.424013	0.232645	0.263111	0.249217	0.264645
Archived r0.45	0.370708	0.201664	0.227298	0.217701	0.219180
Archived recovery	0.366968	0.200468	0.226239	0.216213	0.218749
Archived small	0.535844	0.306309	0.348352	0.318762	0.338282
Fresh control s0	0.464199	0.258842	0.292664	0.278218	0.290396
Fresh control s1	0.464190	0.259101	0.293051	0.278035	0.292689
Fresh control s2	0.464520	0.259413	0.293427	0.278748	0.290167
Fresh CWD s0	0.463144	0.258245	0.292024	0.277702	0.291056
Fresh CWD s1	0.463500	0.258608	0.292497	0.277031	0.290872
Fresh CWD s2	0.463149	0.258157	0.291754	0.277939	0.288906
Fresh large	0.574022	0.333561	0.382573	0.341466	0.378677
Fresh MSE s0	0.464164	0.258620	0.292564	0.278332	0.290253
Fresh MSE s1	0.463939	0.258770	0.292495	0.277766	0.291422
Fresh MSE s2	0.464327	0.259030	0.292875	0.278537	0.290051
Fresh nano	0.463948	0.259863	0.293562	0.280390	0.290070

The released per-image arrays reproduce all global results; the new independent check also recomputes every class and all five image groups for the 11 fresh models, with maximum absolute AP error 0. Every class value, target count and both AP50/AP50:95 group values are retained in the machine-readable metrics. Global and subgroup values have different class populations and should not be subtracted as if they measured the same category average.

S5 Fresh paired illumination differences

Arm	Seed	All 10 pp	Day 9 pp	Night 9 pp	Dawn 9 pp
MSE	0	-0.02217	-0.01002	+0.01140	-0.01429
MSE	1	-0.03312	-0.05559	-0.02686	-0.12673
MSE	2	-0.03828	-0.05519	-0.02100	-0.01159
CWD	0	-0.05969	-0.06401	-0.05157	+0.06597
CWD	1	-0.04931	-0.05539	-0.10038	-0.18167
CWD	2	-0.12561	-0.16726	-0.08087	-0.12606

These batch 8 differences are separate from the batch 32 primary values in S2. All fresh subgroup means are negative; individual signs are mixed for MSE at night and CWD at dawn/dusk. They do not demonstrate a causal illumination effect. All earlier paired subgroup values remain available in the separately labeled earlier-phase analysis.

S6 Protocol and provenance details

The original-format training JSON covers 69,863 images and 1,286,871 positive-area rows; native loading removes one exact duplicate row in 75055858-7d04a650, yielding 1,286,870 targets. Source files remain unchanged. The unknown 137 images and their 1,141 boxes are excluded only from the new phase. Local additional zero-area boxes number 393 in training and 52 in validation. The reconstructed validation has 185,526 boxes. This supports correspondence to the retrieved legacy 2018 mirror, not authentication as Detection 2020 or an official-host archive.

Official COCO checkpoints are downloaded from Ultralytics assets release v8.2.0. Nano and large use 100 epochs, native SGD with Nesterov momentum .937, weight decay .0005, lr .01 to .0001 on the native linear schedule, warm-up 3 epochs and AMP. Nano batch 64 and large batch 16 with nominal-batch 64 accumulation are fixed before training. Native last.pt at epoch 100 is the endpoint; best.pt is not selected. Source hashes, executed scripts and all 100 epoch rows are supplied for both models.

All continuations use 20 epochs, batch 64, one SGD group without Nesterov, momentum .937, weight decay .0005, cosine lr .0001 to .00001, AMP scale 128 and gradient clipping 10. Mosaic, mixup and copy-paste are disabled. The objective is native batch-summed detection loss plus actual batch size times 0.3 times the sum of the three per-scale feature means. CWD temperature is 4. The same numeric coefficient does not equalize gradient strength or optimize both methods. Layers 15/18/21 and learned 1×1 adapters are specified in the code.

KITTI inference uses 640-square letterboxing, rect=False, batch 8, FP32, confidence .001, class-aware NMS .7 and max 300 detections over all ten classes before retaining 0/2. Coordinates are restored to native pixels. Native 2D matching, ignored neighbors, DontCare, difficulty and precision-envelope code are unchanged. Only local paths/frame count, unused 3D/server infrastructure and explicit R40 aggregation differ. Synthetic perfect, missing, false-positive, DontCare, neighboring-class, half-missing, difficulty and low-height cases pass before real inference. The downloaded devkit, patch and compiler/executable hashes allow independent inspection.

S7 Reproduction artifacts

Results commit: c54d758d431ec3dc07927d2a94c63dc4b3031132

Repository: <https://github.com/qkdlsem-maker/yolo-lightweight-optim>

Versioned artifacts: <https://github.com/qkdlsem-maker/yolo-lightweight-optim/releases/tag/ieee-access-reassessment-20260930>

Archive	Content	Bytes
inference_checkpoints_25.zip	25 inference model files	206826850
kitti_predictions_25.tar.gz	187,025 per-image predictions	555057768
bdd_per_image_statistics_27.zip	27 per-image statistics sets	264101626

All three downloaded release assets are checked byte-for-byte against their SHA-256 manifests. Released checkpoint file hashes differ from the original because training-location/history metadata is removed; the manifest records both. Model tensor values and dtypes, module trees and class names match exactly, and all deterministic CPU FP32 raw-forward probe tensors are bitwise equal. This probe is a consistency check, not a replacement for the recorded benchmark inference. Upstream AGPL-3.0 notices remain attached to the inference checkpoints. Underlying images and full source box annotations must be obtained separately under dataset terms.

The repeated BDD validation population, public-mirror provenance, one starting pair per phase, method-specific tuning not performed, correlated KITTI frames, unaudited COCO image overlap, two-category external scope, unequal historical pruning budgets and shared desktop GPU are retained limitations. No adaptive model change follows the external results. The published artifacts enable checking the stated conditional findings; they do not guarantee journal acceptance or deployment safety.